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Summary

Software Engineer at Amazon (Customer Service Data & AI Platform) specializing in **large-scale distributed data infrastructure** and **cloud-native AI/ML infrastructure** for regulated environments. Building **self-service ML tooling** (Python SDKs, CLI frameworks), **AI agents (LangChain, StrandAgents)**, and helping scientists deploy **parallelized models** using GPUs (**Ray, SageMaker, DeepSpeed**) for high-impact org-wide analysis. Expertise in **AWS** (EKS, Kubernetes, EC2, VPC, Kinesis, Lake Formation, Glue), **distributed systems (Ray, Spark, Redis)**, and **ML frameworks (PyTorch, Hugging Face, XGBoost)**. **M.Sc. in Computing Science** with publication in **ICALP**.

Experience

Amazon, Inc

Nov 2022 – Present

Software Development Engineer

Vancouver, BC, Canada

• Project: Multi-Tenant AI/ML Platform for Self-Service Data Access, ML Training, and ETL Job Execution

- * Architected and maintained Amazon Customer Service's internal **AI/ML Platform**, empowering ML scientists with **self-service capabilities** for **ETL workflows, ML training, serving, and productionalization**. The platform enables compliant model development and deployment on sensitive customer data via secure, multi-tenant **VPC-isolated JupyterLab** servers and **Lake Formation**. Achieved platform **adoption metrics** including **200 active workspaces, 50 models deployed to production, and 5 agentic workflows** serving internal customer service applications, significantly reducing time-to-production from 3 weeks to 2 days.
- * **Implemented** the platform's infrastructure provisioning via an **end-to-end CI/CD pipeline**, automating workspace creation (**EKS clusters, EC2, VPCs, IAM roles, security groups**) with one-click deployment, reducing environment setup time from 8 hours to less than 5 minutes, and establishing robust **Kubernetes operations** for cluster lifecycle management.
- * Implemented a scalable runtime for **ML and data job orchestration** using **EKS clusters, Kubernetes operators, and Apache Airflow**.
- * Developed comprehensive developer tooling with three integration layers: Project templates & scaffolding; CLI command suite for managing their **ETL and training jobs**; and a Python SDK with high-level abstractions for distributed training, **PII filtering, and agentic workflows (LangChain, StrandAgents)**.
- * Implemented a **Python SDK** enabling **model and data parallelization** for **large-scale model training, finetuning, and reinforcement learning**, leveraging frameworks like **Ray, DeepSpeed, and PyTorch Lightning** to optimize distributed **GPU** utilization and reduce training times by 40%.
- * Implemented robust **monitoring, logging, and model tracking** solutions for ML pipelines using **MLflow, Weights and Biases, and custom dashboards**; providing experiment traceability, performance visibility, and anomaly detection for deployed models.
- * Implemented specialized tools and workflows enabling ML scientists to write, test, and deploy **AI agents** directly to **Amazon Agent Core**.
- * Implemented resource rate limiting and quota management for **GPU/compute clusters**, with job scheduling and auto-scaling policies ensuring fair multi-tenant access while achieving 90% utilization.
- * Supported ML scientists **debugging** and optimizing models built with **XGBoost, PyTorch, Hugging Face Transformers, JAX, and PyTorch Lightning** – provided tooling for distributed training **troubleshooting** and performance profiling.

• Project: Customer Issue & Escalation Event Pipeline - Real-Time & Analytics

- * Implemented **AI Agents** (using **Claude Agent SDK / Amazon Bedrock Agents**) with an **MCP (Model Context Protocol) server** and custom skills, enabling conversational AI to answer specific business questions directly from processed customer issue data.
- * Built real-time streaming ingestion pipeline using **Amazon Kinesis, SNS/SQS, and Redis** handling **50K+ TPS**, writing customer issue and escalation events to **AWS Lake Formation** Bronze layer with fault-tolerant, exactly-once delivery semantics.
- * Engineered **AWS Glue** jobs for Gold layer transformations processing Bronze data into curated datasets, managing **1.5+ PB** of issue and escalation data optimized for ML model training and business analytics.
- * Implemented data quality checks and schema validation ensuring downstream teams access clean, compliant data for 20 ML workflows across both real-time inference and historical training pipelines.

Delta-X Research

Sep 2021 – Oct 2022

Software Development Engineer

Victoria, BC, Canada

- Developed and maintained a **statistical model** for generating reports on electrical transformer data, using **scikit-learn** and **NumPy** for model development and utilizing various open-source tools and frameworks.
- Built and managed the deployment stack leveraging **Podman** and elements of **Kubernetes** for containerization and orchestration, ensuring consistent development-to-production environments.
- Developed a core **Python library** for connecting to a **range-based sharded database** and integrating with **Durus** (a persistent object system optimized for read-heavy, in-memory caching), simplifying data access and enhancing query performance.

SAP Canada

Oct 2019 – Sep 2021

Software Development Engineer

Vancouver, BC, Canada

- Built and operated secure, high-availability **Java microservices** on **Cloud Foundry**, focusing on **REST APIs** and **OAuth-based** security architectures.
- Developed **full-stack components** including client-side UIs and backend APIs for enterprise applications, enhancing user experience and data interaction.

- Implemented robust authentication and authorization protocols using **OAuth** and managed deployments via **Docker**-based containerization.
- BCI**
Software Development Intern
 - Performed data manipulation and analysis using **R**, **Python**, and **SQL**.
 - Developed interactive data visualization tools in **JavaScript** using the **D3 library**.

Sep 2018 – Aug 2019
Victoria, BC, Canada
- Simon Fraser University**
Research Assistant
 - Authored **"Counting Homomorphisms in Plain Exponential Time"** (published in ICALP 2020), contributing new algorithms for complex graph problems.
 - Developed computational biology models for RNA folding using the Inside-Outside algorithm, optimizing for algorithmic efficiency.

Sep 2016 – April 2019
Burnaby, BC, Canada

Education

- Simon Fraser University**
M.Sc. In Computing Science (GPA 4.2 / 4.30)
Thesis: "Counting homomorphisms in plain exponential time" (Advisor: Prof. Andrei Bulatov)

Sep. 2016 – May. 2019
Burnaby, BC, Canada
- DeepLearning.AI**
Andrew Ng's Deep Learning Specialization Certificate

2024
Verification Link
- AmirKabir University of Technology**
B.Sc. in Software Engineering (GPA 18.30 / 20)

Sep. 2012 – Jul. 2016
Tehran, Iran

Publication

- Counting Homomorphisms in Plain Exponential Time.**
Published in ICALP 2020, A.Dadsetan, A.Bulatov
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Technical Skills & Highlights

- Languages:** Python, Java, Rust, JavaScript, TypeScript, SQL, C
- Cloud & Infrastructure:** AWS (EKS, EC2, VPC, Lambda, Kinesis, Lake Formation, Glue, SageMaker, Bedrock), GCP, Kubernetes, Docker, Terraform, ArgoCD
- ML/AI Frameworks:** PyTorch, Ray, DeepSpeed, Hugging Face, LangChain, XGBoost, JAX, TensorFlow, scikit-learn
- Data & Distributed Systems:** Spark, Airflow, Redis, Kafka, Elasticsearch
- Developer Tools:** Jenkins, GitHub Actions, Android Studio